Phishing

Phishing refers to an attempt to steal sensitive information, typically in the form of usernames, passwords, credits cards number, bank account information, or other important data in order to utilize or sell the stolen information. By disguseing themselves as a reputable source with an urgent request, an attacker lures in the victim in order to trick them, similar to how a fisherman uses bait to catch a fish.

There are several tactics attackers can use to make their phishing attempts more effective:

- Finding, purchasing, or scraping known contact information
- Setting up fake websites and apps that imitate the real ones
- Using techniques like DNS fast fluxing to disguise their hosting servers
- Using domain spoofing and email spoofing to make messages appear legitimate
- Manipulating links so that URLs in phishing messages look correct
- Sending emails from trusted infrastructure that can pass checks and get past spam filters
- Using generative AI to quickly create realistic-sounding and error-free messages

Why Is Phishing a Problem?

Phishing is a significant problem because it’s easy, cheap, and effective, making them widespread cyber-attacks. Victims of phishing scams may end up with malware infections (including ransomware), identity theft, and data loss. Unlike traditional security threats that target technical vulnerabilities, phishing directly exploits the human element of security by often the most vulnerable link in an organization’s security as they penetrate into the system through a person clicking on link or downloading the malicious software unknowingly which executes directly in the systems mainframe and spreads if the systems are connected or spread fraudulent emails to other connected devices through mail in the name of the infected user. Phishing is also used in modern times to gain access to social media accounts, which could be used as mean to steal sensitive information , black mail or cyber bullying .

How do Hackers make themselves untraceable?

Phishing attackers use a combination of technical “smoke and mirrors” and psychological triggers to hide their tracks. Tracing an email usually involves getting its IP address. Attackers ensure this IP doesn’t point them out. With help of VPN and Tor these attackers make up their fake addresses to deceive the tracker. They host their malicious links and fake login pages on specialized web hosts that ignore legal take-down notices and law enforcement requests private cloud hosting places are also a good space for attackers. By putting the malicious link inside a QR code, attackers bypass traditional URL scanners that only look at plain text links, they also use virtual numbers which do not give out real information where the attacker is based at.

A Real-Life Incident:

Source: https://www.hempsteadny.gov/635/Famous-Phishing-Incidents-from-History

The Nordea Bank Incident

In 2007, Swedish bank Nordea faced a loss of 7 million kronor when phishers managed to send fraudulent emails out to bank customers, which requested them to install the "haxdoor" Trojan disguised as anti-spam software. It was called as the "biggest ever online bank heist" by digital security company McAfee, Nordea Bank customers were hit with phishing emails containing Trojan viruses that installed a keylogger *(A malware which records every keystroke of the victim and sends those key strokes to its main server)* into the victims' computers and directed them to a fake bank website where hackers intercepted the customers login credentials. While the exact blame can't be reliably placed, it is worth noting that most customers failed to have a running antivirus installed on their machines.